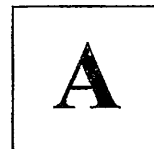


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***B.Tech. Degree III Semester Supplementary Examination
November 2020***

**ME 15-1304 FLUID MECHANICS
(2015 Scheme)**

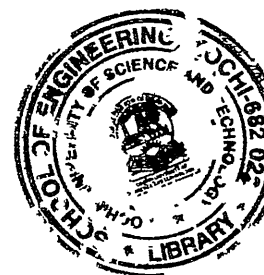
Time: 3 Hours

Maximum Marks. 60

**PART A
(Answer *ALL* questions)**

(10 × 2 = 20)

- I. (a) Distinguish between the following:
 - (i) Ideal fluid and Real fluid
 - (ii) Poise and stokes
- (b) Define: (i) Total pressure (ii) Centre of pressure.
- (c) Define path line, streak line and stream line. For what type of flow these lines are identical.
- (d) What is piezometer? What are its limitations?
- (e) Differentiate between laminar and turbulent flows.
- (f) Explain the causes of major and minor energy losses in pipe flows.
- (g) Explain the concept of circulation and vorticity.
- (h) An idealized flow is given by $V = 2x^3i - 3x^2yj$. Check the possibility of flow. State whether the flow is steady or unsteady.
- (i) What do you mean by boundary layer thickness?
- (j) What is the effect of pressure gradient on boundary layer separation?



PART B

(4 × 10 = 40)

- II. (a) A plate weighing 150N and measuring 0.8 m x 0.8 m slides down an inclined plane over an oil film of 1.2 mm thickness for an inclination of 30° and a velocity of 0.2 m/s. Compute the dynamic viscosity of the fluid. (6)
- (b) Describe the use of Reynolds transport equation to formulate fluid mechanics problems. (4)

OR

- III. A rectangular pontoon 10 m long 7 m broad and 2.5 m deep weights 686.7KN. It carries on its upper deck an empty boiler of 5 m diameter weighing 588.6 KN. The centre of gravity of the boiler and the pontoon are at their respective centers along a vertical line. Find the metacentric height and check the stability of the body. Weight density of sea water is 10.104 KN/m^3 . (10)

(P.T.O.)

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- IV. (a) Derive Euler's equation and hence deduce the expression for Bernoulli's Equation. State the assumptions made for such derivation. (6)

- (b) What are the applications and limitations of Bernoulli's equation? (4)

OR

- V. (a) Glycerine flows at a velocity of 6 m/s in a 10 cm diameter pipe. Dynamic viscosity and density of glycerine is assumed as 1.50 Pa.s and 1260 kg/m³ respectively. Estimate: (6)

(i) The boundary shear stress in the pipe due to the flow.

(ii) Head loss in a length of 12 m of pipe.

(iii) Power developed by the flow in a distance of 12 m

- (b) Explain Moody's chart. What is its use in pipe flow? (4)

- VI. (a) The velocity potential function for a two dimensional flow is $\phi = x(2y-1)$ at a point P (4,5). Determine: (6)

(i) The velocity

(ii) The value of stream function.

- (b) What is Magnus effect? Explain. (4)

OR

- VII. A uniform flow with a velocity of 3 m/s is flowing over a plane source of strength 30 m²/s. The uniform flow and source flow are in the same plane. A point P is situated in the flow field. The distance of the point P from the source is 0.5 m and it is at an angle of 30° to the uniform flow. Determine: (10)

(i) stream function at point P, (ii) resultant velocity of flow at P (iii) location of stagnation point from the source.

- VIII. Workout the following boundary layer parameters for the velocity (10)

distribution prescribed by $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{\frac{1}{7}}$

(i) Displacement thickness

(ii) Momentum thickness

(iii) Energy thickness

(iv) Shape factor

Further evaluate the energy loss due to boundary layer if at a particular section the boundary layer thickness is 3 cm and the free stream velocity is 12 m/s. if the discharge through the boundary layer region is 5 m³/s per metre width, express the energy loss in terms of metre of head. Take density = 1.2 kg/m³.

OR

- IX. (a) Derive an expression for Von-karman momentum integral equation (6)

- (b) Write down different methods to prevent boundary layer separation (4)
